# WASP-10b

**R-0814**

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_WASP-10_181002` · created 2026-07-28T12:53:53+00:00

**PRACTICE / VALIDATION RUN — not submitted to any science database**

## Results

| | |
|---|---|
| Mid-Transit Time (Tmid) | 2458393.8636 +/- 0.0041 BJD_TDB |
| Ratio of Planet to Stellar Radius (Rp/R*) | 0.205 +/- 0.061 |
| Transit depth (Rp/Rs)^2 | 4.21 +/- 2.49 [%] |
| Orbital Inclination (inc) | 85.88 +/- 1.91 |
| Airmass coefficient 1 (a1) | 8.27 +/- 5.89 |
| Airmass coefficient 2 (a2) | -1.67 +/- 0.62 |
| Scatter in the residuals of the lightcurve fit is | 56.79 % |
| Best Comparison Star | #2 - [116, 171] |
| Optimal Aperture | 2.88 |
| Optimal Annulus | 6.75 |
| Transit Duration (day) | 0.063 +/- 0.033 |

## Accuracy vs published ephemeris (ExoClock/NEA)

| | |
|---|---|
| Rp/R■ fit vs published | 0.205 ± 0.061 vs 0.1592 (deviation 0.75σ) |
| Duration fit vs published | 0.063 d vs 0.0946 d (deviation 0.96σ) |
| Mid-time O–C | -6.5 min |
| Depth SNR | 3.4 |
| Residual scatter | 56.79 % |

## Light curve

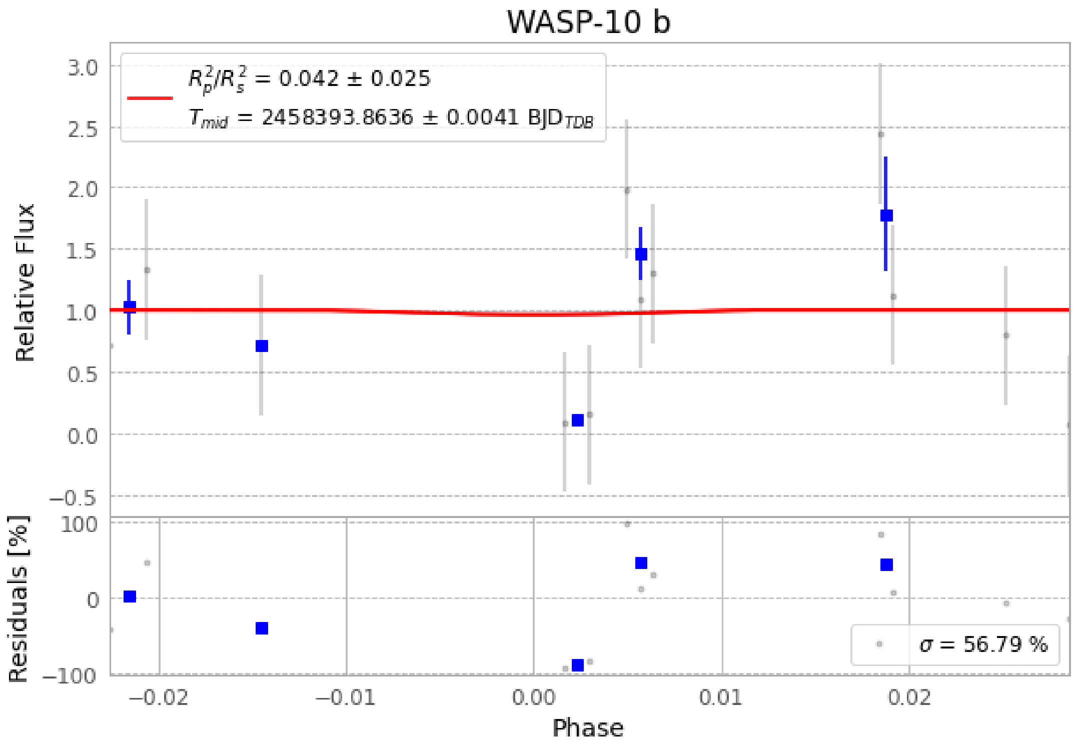


Figure 1 — FinalLightCurve_WASP-10 b_2018-10-01.png (SHA-256 ce41aec9a5a5fcb3…)

## Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [134, 225], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 93c4f078d6128ef5…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9`

## Data provenance

84 FITS frames (55 MB), `WASP-10181002064512.FITS` → `WASP-10181002105412.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-10-181002.log (55f42511a88b…), FinalLightCurve_WASP-10 b_2018-10-01.png (ce41aec9a5a5…), FinalLightCurve_WASP-10 b_2018-10-01.csv (a608cb875806…)

## Compute resources

Wall time 200.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)